The International Marine Contractors Association

Battery Packs in Pressure Housings

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The International Marine Contractors Association (IMCA) is the international trade association representing offshore, marine and underwater engineering companies.

IMCA promotes improvements in quality, health, safety, environmental and technical standards through the publication of information notes, codes of practice and by other appropriate means.

Members are self-regulating through the adoption of IMCA guidelines as appropriate. They commit to act as responsible members by following relevant guidelines and being willing to be audited against compliance with them by their clients.

There are two core committees that relate to all members:

- Safety, Environment & Legislation
- Training, Certification & Personnel Competence

The Association is organised through four distinct divisions, each covering a specific area of members' interests: Diving, Marine, Offshore Survey, Remote Systems & ROV.

There are also four regional sections which facilitate work on issues affecting members in their local geographic area – Americas Deepwater, Asia-Pacific, Europe & Africa and Middle East & India.

IMCA Diving Division

The Diving Division is concerned with all aspects of the equipment, operations and personnel of offshore diving operations, including atmospheric diving systems.

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Battery Packs in Pressure Housings

1 OBJECTIVES

The objectives are to:

a) draw attention to the problems which can arise with battery packs when used in pressure housings.

b) provide guidance on their fitting and operation.

2 APPLICATION

This guidance is applicable in any geographic area in addition to national regulations which must always be adhered to.

3 BACKGROUND

An incident occurred involving a bell external battery assembly when the bursting disc fitted to the bottom of the pressure housing ruptured discharging the contents with considerable force. It was, in effect, a small explosion resulting in a significant gas and fume discharge with a high potential for serious injury to personnel.

4 CAUSAL FACTORS

4.1 The immediate cause was found to be an electrical short-circuit. This resulted in a power drain on the battery leading to heat generation which promoted further cell breakdown, further short-circuiting, overheating and pressure build up in the casing followed eventually by violent venting from the battery casing via the bursting disc.

4.2 It was not possible to establish beyond doubt:

a) the cause of the initial short circuit. It is known however that the battery was 10 years old and was not protected by a shunt diode.

b) whether there had been insulation on the negative lead.

5 RECOMMENDATIONS

The following criteria should be observed:

1. Batteries should always be used in accordance with manufacturer's recommendations.
2. Installation, maintenance and servicing of battery packs in pressure housings should only be performed by a competent person.
3. Battery terminals/leads must be adequately insulated to protect against short circuit.

4. Periodic examination, testing and renewal of the cells as necessary should be included within the planned maintenance system.
5. As a general rule shunt diodes should be provided across each cell of a primary battery to avoid the possibility of polarity-reversal occurring in any cell under discharge conditions.
6. The battery housing must be fitted with an appropriate pressure relief device.
7. Batteries should be charged in line with the guidance given in AODC 054 "Prevention of Explosions during Battery Charging in Relation to Diving Systems".
8. The battery casing should not be opened in a confined space and should be fully vented. Consideration should be given to purging the housing with air prior to opening. Suitable eye and respiratory protection should be worn in case of a sudden release of pressure and noxious fumes which may be harmful if inhaled.
9. Lead acid batteries should not be used in a hyperbaric environment.
10. Battery packs being put into storage should be suitably prepared for the length of time envisaged.